

Finite self-direct-sums for a regular C-type frequently hypercyclic class

New partial result for a question in arXiv:2007.12647

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Status. **New partial result; the unrestricted question remains open.** For the principal regular $C_{+,1}$ -type class, including the known chaotic frequently hypercyclic example which is not hereditarily frequently hypercyclic, frequent hypercyclicity survives every finite self-direct-sum. Thus that central candidate family cannot yield a counterexample to the question whether $T \oplus T$ is frequently hypercyclic whenever T is.

Source question and later boundary

Gilmore's survey [1] asks whether frequent hypercyclicity of a bounded operator T implies frequent hypercyclicity of $T \oplus T$. Here T is frequently hypercyclic if some vector visits every nonempty open set at a set of times of positive lower density.

The question is still stated as open in [3]. That paper introduces hereditary frequent hypercyclicity, a stronger property which does imply the desired conclusion, and then constructs a chaotic frequently hypercyclic $C_{+,1}$ -type operator which is not hereditary. The construction therefore tests whether failure of heredity might also break the self-direct-sum.

The C-type parameters

We use the notation of [2]. At level k , the relevant $C_{+,1}$ -type blocks have common length Δ_k , feedback magnitude $2^{-\tau_k}$, and internal weights

$$w_i^{(k)} = \begin{cases} 2, & 1 \leq i \leq \delta_k, \\ 1, & \delta_k < i < \Delta_k. \end{cases}$$

The block map sends each sufficiently high block to an earlier block, and every finitely supported vector is periodic. The quantity

$$\eta := \limsup_{k \rightarrow \infty} \frac{\delta_k - \tau_k}{\Delta_k}$$

measures the positive amplification margin available for periodic orbit shadowing.

Theorem 1 (finite-self-sum theorem). *Let $1 \leq p < \infty$, and let T be a $C_{+,1}$ -type operator on $\ell_p(\mathbb{N})$ with $\eta > 0$. Then, for every integer $r \geq 1$,*

$$T^{\oplus r} := \underbrace{T \oplus \cdots \oplus T}_{r \text{ copies}}$$

is frequently hypercyclic.

The same proof works under the more general pair of weight-product hypotheses in the periodic-shadowing theorem of [2]; the displayed limsup is a convenient readable sufficient condition.

Idea of the proof

The one-copy proof in [2, Thms. 5.31 and 6.9] does more than produce one frequently hypercyclic vector. Given a finitely supported periodic target x , it chooses one high block level k , one transfer time m , and one common period $d = 2\Delta_k$. A correction $z = L_{k,m}x$, linear in x , is small along a fixed positive fraction of its period and, after time m , shadows the same fraction of the orbit of x .

For finitely many targets $x_1, \dots, x_r$, choose a level above all their supports and make the same weight-product lower bound large enough for the largest of their norms. The same k, m, d then works in every coordinate. The tuple $(L_{k,m}x_1, \dots, L_{k,m}x_r)$ supplies exactly the periodic shadowing condition for $T^{\oplus r}$.

Proof

Let $X_0 = c_{00} \subset \ell_p(\mathbb{N})$. It is dense, T -invariant, and consists of periodic vectors. The frequent-hypercyclicity criterion used in [2] has the following operative form. For some $\rho > 0$, for every $x \in X_0$, tolerance $\varepsilon > 0$, and previously fixed finite periodic data, one finds a periodic $z \in X_0$ and integers $m, d \geq 1$ such that d is a common admissible period and

$$\|T^j z\| < \varepsilon, \quad \|T^{m+j} z - T^j x\| < \varepsilon \quad (0 \leq j \leq \rho d). \quad (1)$$

This condition implies frequent hypercyclicity by concatenating the positive proportions of periodic shadowing times.

For $C_{+,1}$ -type operators, the proof of the periodic-shadowing theorem constructs z in descendant blocks at a level k . For fixed k, m , its formula (displayed in the proof of Theorem 6.7, to which the proof of Theorem 6.9 explicitly refers) is linear in the coefficients of x . If the support of x lies below level k_0 , both errors in (1) are bounded by

$$C^{-1} A^{\Delta_{k_0}} \|x\|,$$

where A is independent of x , once the two level- k weight products are at least C . The correction has common period $d = 2\Delta_k$, and the source's divisibility choice accommodates the earlier periodic data.

We verify that arbitrarily large common bounds C are available. Choose $0 < a < \eta/2$. For arbitrarily large k , $\delta_k - \tau_k > 2a\Delta_k$. Taking k larger still makes $a\Delta_k > \log_2 C$. With the source proof's choice $m = \Delta_k - 1$, the first relevant product is

$$2^{-\tau_k} \prod_{i=1}^{\Delta_k-1} w_i^{(k)} = 2^{\delta_k - \tau_k} > C,$$

and, uniformly for $0 \leq j \leq a\Delta_k$, the second is at least

$$2^{\delta_k - \tau_k - j} > 2^{a\Delta_k} > C.$$

These are precisely the amplification bounds used to obtain (1): since $d = 2\Delta_k$, take $\rho = a/2$, so that $\rho d = a\Delta_k$.

Now fix r , use the max norm on the finite product, and take $\mathbf{x} = (x_1, \dots, x_r) \in X_0^r$. Choose one k_0 above all supports and choose C so large that $C^{-1} A^{\Delta_{k_0}} \max_i \|x_i\| < \varepsilon$. Use the same good k, m in all coordinates and put

$$\mathbf{z} = (L_{k,m}x_1, \dots, L_{k,m}x_r).$$

Every coordinate has the same admissible period d , and the coordinatewise estimates give (1) for $T^{\oplus r}$ and $\mathbf{x}$. Since X_0^r is dense, invariant, and periodic, the same criterion proves that $T^{\oplus r}$ is frequently hypercyclic.

Application to the non-hereditary example

In the example of [3],

$$\delta_k = \frac{\Delta_k}{4}, \quad \tau_k = \frac{\Delta_k}{8}, \quad \frac{\delta_k - \tau_k}{\Delta_k} = \frac{1}{8}.$$

The block lengths are then chosen sufficiently rapidly to prove that T is not frequently hypercyclic along one positive-lower-density set. Hence T is chaotic and frequently hypercyclic but not hereditarily frequently hypercyclic. Theorem 1 nevertheless gives $T^{\oplus r} \in \text{FHC}$ for every finite r .

There is also a broader regular-class corollary. Under the monotonicity and summability hypotheses in the $C_{+,1}$ equivalence theorem of [2], and assuming $\limsup \tau_k/\delta_k < 1$, frequent hypercyclicity is equivalent to $\limsup \delta_k/\Delta_k > 0$. It therefore forces $\eta > 0$, so every frequently hypercyclic operator in that regular class has frequently hypercyclic finite self-sums.

Scope and failed full upgrades

Route	Outcome
Regular $C_{+,1}$ candidates	Ruled out by the theorem above.
Single bounded-below weighted shifts	Known finite-self-sum stability; no counterexample.
Two-shift bad pair encoded into one shift	The incompatible support mechanism vanishes for identical copies.
Common extension of the bad pair	No frequently hypercyclic extension with the two required quotient maps was found.
Invariant-measure products	Frequent hypercyclicity alone supplies neither a full-support weakly mixing measure nor an adequate product theorem.
Arbitrary C-type operators	No positive shadowing margin follows from frequent hypercyclicity outside the regular subclass.

Thus this is a substantive elimination theorem, not a solution of the unrestricted question. A full counterexample must leave both the standard weighted-shift mechanism and the synchronized periodic-shadowing C -type regime, or find a new way to defeat simultaneous positive lower densities.

Novelty and verification. Exact arXiv-id, phrase, and core-keyword searches through 11 August 2026 found the survey question, the 2024 still-open statement, and the two-different-shift counterexample, but no prior statement of the finite-self-sum consequence above. The proof is an explicit coordinatewise audit of the published periodic-shadowing estimates. Human review should focus on the common-period divisibility step and the uniformity of the constants in the transfer formula.

References

- [1] E. Gilmore, *Linear Dynamical Systems*, arXiv:2007.12647.
- [2] S. Grivaux, E. Matheron, and Q. Menet, *Linear dynamical systems on Hilbert spaces: typical properties and explicit examples*, arXiv:1703.01854. See Theorems 5.31, 6.9, 6.17, and 6.18, together with the transfer formula in the proof of Theorem 6.7.
- [3] F. Bayart, S. Grivaux, E. Matheron, and Q. Menet, *Hereditarily frequently hypercyclic operators and disjoint frequent hypercyclicity*, arXiv:2409.07103. See the still-open self-direct-sum question on p. 2 and the chaotic non-hereditarily frequently hypercyclic $C_{+,1}$ example on p. 25.